

TREMORTRACE · REQUIREMENTS

TremorTrace requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	TRT-REQ-001	Version	0.2
Status	Draft	Date	2026-09-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

TremorTrace requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Measure tremor frequency across the clinically relevant band	3 to 15 Hz, resolution 0.2 Hz or better	Calculation of sampling and window length; later bench shaker test
R2	Measure tremor amplitude	Acceleration RMS and angular velocity RMS in the tremor band, reported per window	Calculation; later comparison against a reference sensor
R3	Sample motion fast enough	6-axis IMU at 100 Hz or more	Datasheet and firmware configuration
R4	Run a full waking day and more	3 days or more between charges at 16 h per day of wear	Power budget calculation
R5	Store a day of results on the band	7 days of 10 s window summaries without a phone	Storage calculation
R6	Comfortable for all-day wear	Mass 30 g or less including strap; pod no larger than 45 x 35 x 14 mm	Massing model, then weighing
R7	Skin-safe and wearable	Silicone or textile strap; no exposed electronics; splash resistant (IP54 target)	Design review
R8	Produce a daily tremor profile	Open notebook turning window summaries into a daily chart of tremor time, frequency and amplitude	Run on sample data
R9	Protect health data	Data stored locally and exported only by the owner; no cloud dependency	Design review
R10	Low cost and buildable	Parts cost \$150 or less per unit; no custom PCB required for the first build	Priced BOM
R11	Clear status	LED or phone app shows battery, recording and sync state	Design review

■ Assumptions

- Parkinsonian rest tremor is typically about 4 to 6 Hz and essential tremor about 4 to 12 Hz; a 3 to 15 Hz band covers both with margin.
- Wear time of 16 h per day and nightly charging is acceptable, but a multi-day battery is preferred so a missed charge does not lose a day.